

# BCA Specializations – Semester-wise Practical & Lab Subjects Analysis

This report provides a semester-wise analysis of BCA specializations under Bengaluru City University (SEP 2024 Scheme). The document highlights practical-oriented subjects, laboratory courses, and specialization-focused papers for the following streams: BCA Full Stack Development BCA Artificial Intelligence & Machine Learning BCA Data Science

## BCA Full Stack Development

| Semester     | Practical / Lab / Specialization Subjects                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semester I   | <ul style="list-style-type: none"><li>• Problem Solving Technique Lab</li><li>• Computer Architecture Lab</li><li>• Office Automation Tools</li></ul>                       |
| Semester II  | <ul style="list-style-type: none"><li>• Data Structure Lab</li><li>• Operating Systems Lab</li><li>• Linux and Shell Programming Lab</li></ul>                              |
| Semester III | <ul style="list-style-type: none"><li>• Database Management System Lab</li><li>• Artificial Intelligence Lab using Python</li><li>• Elective: Web Programming – I</li></ul> |
| Semester IV  | <ul style="list-style-type: none"><li>• Computer Networks Lab</li><li>• Design and Analysis of Algorithms Lab</li><li>• Elective: Web Programming – II</li></ul>            |
| Semester V   | <ul style="list-style-type: none"><li>• Frontend Design Lab</li><li>• Backend Development Lab</li><li>• Cloud Computing</li></ul>                                           |
| Semester VI  | <ul style="list-style-type: none"><li>• Mobile Application Development</li><li>• Software Project Management</li><li>• Project Work</li></ul>                               |

## BCA Artificial Intelligence & Machine Learning

| Semester     | Practical / Lab / Specialization Subjects                                                                                                                                   |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semester I   | <ul style="list-style-type: none"><li>• Problem Solving Technique Lab</li><li>• Computer Architecture Lab</li><li>• Office Automation Tools</li></ul>                       |
| Semester II  | <ul style="list-style-type: none"><li>• Data Structure Lab</li><li>• Operating Systems Lab</li><li>• Linux and Shell Programming Lab</li></ul>                              |
| Semester III | <ul style="list-style-type: none"><li>• Database Management System Lab</li><li>• Artificial Intelligence Lab using Python</li><li>• Elective: Feature Engineering</li></ul> |

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|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semester IV | <ul style="list-style-type: none"> <li>• Computer Networks Lab</li> <li>• Design and Analysis of Algorithms Lab</li> <li>• Elective: Introduction to ML</li> </ul>                                                                    |
| Semester V  | <ul style="list-style-type: none"> <li>• ML &amp; Neural Network</li> <li>• Digital Image Processing</li> <li>• ML &amp; Neural Network Lab</li> <li>• Digital Image Processing Lab</li> <li>• Natural Language Processing</li> </ul> |
| Semester VI | <ul style="list-style-type: none"> <li>• Deep Learning for Computer Vision</li> <li>• Predictive Analysis</li> <li>• Project Work</li> </ul>                                                                                          |

## BCA Data Science

| Semester     | Practical / Lab / Specialization Subjects                                                                                                                                                                    |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semester I   | <ul style="list-style-type: none"> <li>• Problem Solving Technique Lab</li> <li>• Computer Architecture Lab</li> <li>• Office Automation Tools</li> </ul>                                                    |
| Semester II  | <ul style="list-style-type: none"> <li>• Data Structure Lab</li> <li>• Operating Systems Lab</li> <li>• Linux and Shell Programming Lab</li> </ul>                                                           |
| Semester III | <ul style="list-style-type: none"> <li>• Database Management System Lab</li> <li>• Artificial Intelligence Lab using Python</li> <li>• Elective: Basics of Data Analytics using Spreadsheet</li> </ul>       |
| Semester IV  | <ul style="list-style-type: none"> <li>• Computer Networks Lab</li> <li>• Design and Analysis of Algorithms Lab</li> <li>• Elective: Data Visualization</li> </ul>                                           |
| Semester V   | <ul style="list-style-type: none"> <li>• Introduction to Data Science</li> <li>• Time Series Analysis</li> <li>• Machine Learning</li> <li>• Data Science Lab</li> <li>• Time Series Analysis Lab</li> </ul> |
| Semester VI  | <ul style="list-style-type: none"> <li>• Big Data Analytics</li> <li>• Exploratory Data Analysis</li> <li>• Project Work</li> </ul>                                                                          |

## Practical Subject Analysis

The BCA curriculum emphasizes hands-on practical learning from the first semester onwards. Core laboratory subjects such as Programming Labs, Computer Architecture Labs, Database Labs, Operating Systems Labs, and Networking Labs provide students with strong implementation skills. **Full Stack Development:** Focuses on frontend, backend, web programming, cloud computing, and mobile applications. **AI & ML:** Emphasizes machine learning, neural networks, deep learning, image processing, and predictive systems. **Data Science:** Concentrates on analytics, visualization, machine learning, big data, and exploratory data analysis. The specialization subjects from Semester III onwards clearly align students toward industry-oriented career paths.